

Measuring progress in pregnancy planning and preconception health



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As efforts to support pregnancy planning and improve preconception health are increasing at scale, appropriate systems to monitor progress are required. Despite developments in a few countries, no surveillance systems currently in operation are using a comprehensive set of indicators for monitoring preconception health. This Review describes relevant indicators, reflecting both system-level and individual-level factors, that can be drawn from routine data sources to form the basis for developing new surveillance systems. We present a new framework for national and international surveillance that incorporates, for the first time, community perspectives on the factors that matter most before pregnancy and parenthood. Finally, we describe an international collaboration working towards a core set of indicators that can be compared across low-income, middle-income, and high-income countries, and discuss future directions to enhance and expand international monitoring of pregnancy planning and preconception health.

Introduction

Interventions and policies to improve pregnancy planning and preconception health are increasingly being developed and implemented globally through health systems and public health programmes.¹⁻⁴ This development reflects growing evidence about the benefits of optimising health before and between pregnancies, not only to improve pregnancy and birth outcomes, but also to reduce intergenerational inequalities and chronic disease risk.⁵⁻⁷

Despite increasing efforts to improve preconception health—alongside improvements in antenatal care and public health measures, such as immunisation programmes—many countries have seen a worsening in maternal and child health outcomes in the past decade. Maternal and perinatal mortality rates have increased,⁸ as has the proportion of children living with obesity.⁹ These intergenerational adverse outcomes are partly fuelled by more people entering pregnancy at an older age, or with obesity, diabetes, or mental illness.¹⁰⁻¹² Substantial inequalities exist in these outcomes and their underlying factors, many of which originate before conception.¹³⁻¹⁵ Addressing these challenges requires a dual approach: (1) effective public health policies focusing on the causes of inequalities and addressing structural barriers to ensure equal access to resources and support (ie, upstream factors), and (2) high-quality health care to support individuals planning pregnancy to optimise health behaviours and conditions.^{16,17} Equally important is the need for greater public awareness of the value of planning for a healthy pregnancy.¹⁸

Further investment in reducing inequalities in the population's preconception health now requires comprehensive monitoring of progress to assess success and inform targeted investments (figure 1). Routine surveillance with a comprehensive set of indicators is not currently in operation at any regional, national, or international level. However, data on relevant indicators that might already be collected routinely, such as

smoking, immunisation, and chronic health conditions, can be described and used as the basis for developing a more comprehensive surveillance system for preconception health. Surveillance should monitor indicators that matter to the community, reflect the effect of interventions, and help policy makers with decision making.

In the context of pregnancy planning and preconception health, we define surveillance as the ongoing systematic

Key messages

- The growing number of preconception health interventions and policies being developed and implemented globally require monitoring to guide future targeted investments.
- Changes in pregnancy planning and preconception health should be measured at both system and individual levels by use of robust indicators that reflect the needs of communities, practitioners, and policy makers.
- Many indicators of pregnancy planning and preconception health can already be measured through a range of existing data sources.
- An internationally agreed core set of priority preconception health indicators does not currently exist but is being developed to facilitate global comparisons of progress and identification of effective preconception health strategies.
- Some countries are establishing systems for routine surveillance of pregnancy planning and preconception health, which can provide case studies to follow or adapt.
- Measuring and monitoring preconception health is a prime opportunity to support governments, health systems, and policy makers to tackle inequalities in intergenerational issues in population health.
- When proposing a new surveillance system to monitor preconception health (or updating an existing one), costs and benefits should be outlined in a business case.

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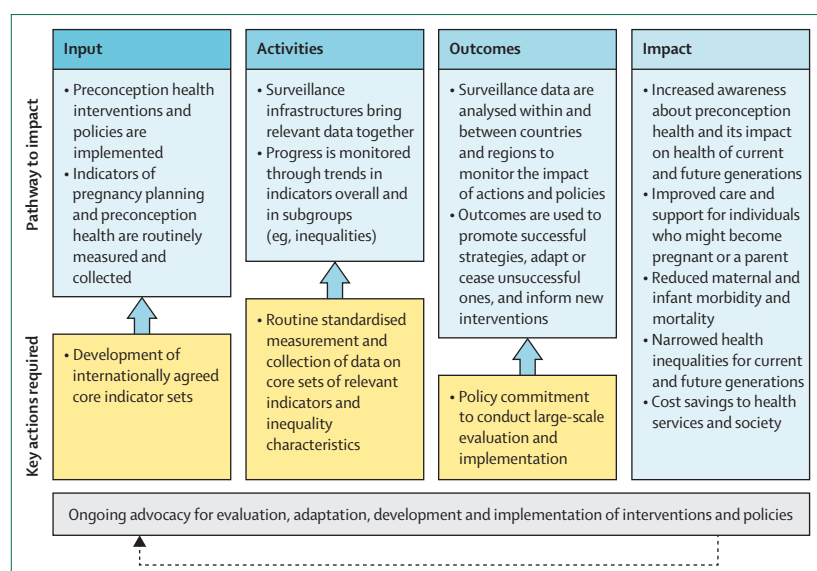


Figure 1: Pathway to impact and key actions required for local, national, and international measurement and surveillance of pregnancy planning and preconception health

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collection, analysis, and interpretation of preconception health indicators at population and system levels to inform the planning, implementation, and evaluation of efforts to optimise health before and between pregnancies, improve pregnancy outcomes, and reduce health inequalities in parents and the next generation.

Preconception health indicators can be defined as medical, behavioural, and psychosocial risk factors or exposures, and broader determinants of health, that can affect potential future pregnancies in people of reproductive age (irrespective of sex and gender).¹⁹

In this Review, we describe indicators of pregnancy planning and preconception health that are relevant to people of reproductive age, whatever their pregnancy intention. We propose a new framework for national and international surveillance that incorporates community perspectives on the factors that matter most before pregnancy and parenthood. We present examples of national surveillance infrastructures and discuss future directions to enhance and expand international monitoring of pregnancy planning and preconception health.

Measures of pregnancy planning and preconception health

At first glance, pregnancy planning and preconception health might appear to be overlooked by routine data collection systems, unlike, for example, maternal health during pregnancy. Pregnancy planning and conception are typically considered private matters and, unless there is difficulty conceiving, the preconception period often passes without attention from health, education, or other routine services. The topics of pregnancy intention and provision of preconception care are rarely discussed,

let alone recorded, in national data sources.^{19,20} However, when the full scope of preconception health and care is considered, many relevant indicators of pregnancy planning and preconception health are measured for various purposes and can be found across a range of data sources (figure 2).

Preconception health indicators can be measured at the individual or system level. Indicators that describe individuals' behaviours and health conditions are most commonly used or proposed for surveillance. Examples include taking a folic acid supplement before conception^{21–24} and prevalence of anaemia and obesity in women of reproductive age.^{10,25} Monitoring these indicators is important because they reflect people's health status and can directly affect pregnancy and birth outcomes. However, focusing solely on individual-level indicators could imply that negative health outcomes result from individuals' poor behaviours and choices. System-level indicators, which reflect broader social, economic, and environmental conditions that influence an individual's ability to plan and prepare for pregnancy, are therefore also needed.

System-level indicators

Currently, system-level indicators are not well developed or described in the context of preconception health, but have been successfully developed in other areas, such as monitoring progress of the Sustainable Development Goals²⁶ and evaluating childhood obesity interventions.²⁷ These indicators could be adapted, and new ones developed, as counterparts to individual-level factors in monitoring optimal support for preconception health (table 1). For example, fortification of staple foods (eg, flour or cereals) with folic acid is a system-level indicator, and taking folic acid supplements an individual-level indicator, of neural tube defects prevention. Individuals' ability to consume a healthy diet and meet physical activity guidelines could be assessed through system-level indicators related to their food environment (eg, availability and affordability of healthy foods) and community infrastructure (eg, proximity and accessibility of parks and gyms).¹⁶

Individual-level indicators

Pregnancy planning

Unplanned pregnancy is a predictor of poor pregnancy and postpartum outcomes such as preterm birth and postpartum depression, independent of sociodemographic and health characteristics.^{37,38} Although the initial categorisation of pregnancies in the USA as intended, mistimed, or unwanted had a profound influence on the framing of survey questions and family planning policy,^{39,40} advances in the understanding of reproductive experience and the development of more robust methodologies have led to new measurement approaches.

A prime example is the London Measure of Unplanned Pregnancy (LMUP). The LMUP is a six-item psychometric

measure that is highly acceptable to users and easy to complete.^{41,42} The LMUP assesses the level of planning for current or recent pregnancies (ie, within the past year). As of 2025, the LMUP has been evaluated in 20 language versions across 17 low-income, middle-income, and high-income countries,⁴³ and partner versions have been developed.⁴⁴⁻⁴⁶ In the UK, the LMUP serves as the standard national tool for assessing unplanned pregnancy,⁴⁷ and studies in antenatal services in the UK⁴⁸ and Australia^{49,50} have shown acceptability in women and midwives. Beyond clinical applications, the LMUP offers opportunities for ongoing public health surveillance of unplanned pregnancy, particularly if integrated into routine health data collection systems across services that ultimately capture all pregnancies (including early losses and abortions).

Although the LMUP is increasingly used worldwide, assessment of individuals' current thinking about a future pregnancy remains less well developed.⁵¹ Validated measures, such as the Desire to Avoid Pregnancy scale,⁵²⁻⁵⁶ are emerging and could help to quantify the need for preconception care services. However, pregnancy planning does not necessarily imply better pregnancy preparation (eg, better health and health behaviours in women and partners who are planning pregnancy).⁵⁷⁻⁶⁴ Therefore, system-level and individual-level indicators that reflect preparation for pregnancy are also needed.

Preconception health

Numerous indicators can be used to describe and monitor preconception health at a population level (table 1). Our review identified 65 indicators across 12 topic areas and domains, including: wider determinants of health, health care, emotional and social health and support, reproductive health and family planning, health behaviours and weight, environmental exposures, cervical screening, immunisation and infections, mental health conditions, physical health conditions, medication use, and genetic risk.¹⁹

Many preconception health indicators are relevant to public health in general (eg, the proportion of people who smoke or have micronutrient deficiencies), regardless of people's plans for future pregnancy and parenthood.^{5,65} For those who do become pregnant (planned or unplanned), optimal preconception health can reduce risks during and after pregnancy for mothers and their children, including conditions such as type 2 diabetes and preterm birth. Certain indicators, such as taking a folic acid supplement and adjusting teratogenic medications (eg, valproate, warfarin, and isotretinoin), are specifically relevant to the preconception period, and rely on pregnancy planning. Other indicators can be used to describe and monitor national policies and laws (eg, effective contraceptive methods that are covered by insurance plans or available at zero or reduced cost) and societal values (eg, family pressure related to having children or not). Changes in these indicators could

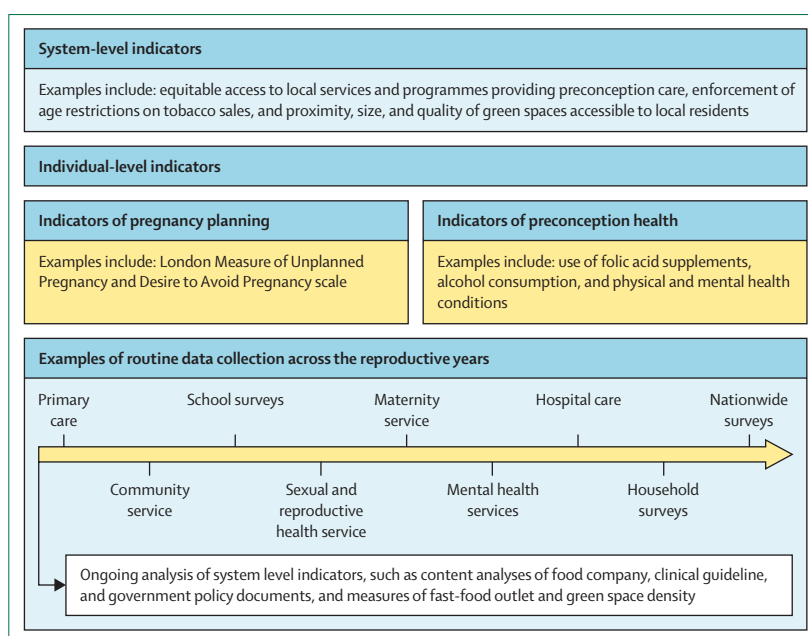


Figure 2: Framework for routine data collection of system-level and individual-level indicators

explain individuals' pregnancy planning and preconception health, and any associated inequalities.

Public involvement to identify relevant and important indicators

Our review of indicators of preconception health was based on academic literature, policy statements, and clinical guidelines.¹⁹ The extent to which these indicators reflect what the public find important before pregnancy and parenthood has been largely unexplored in the context of surveillance. To address this gap, we conducted a series of surveys and interviews with people of reproductive age (ie, age 18–50 years) across middle-income and high-income countries as part of a public involvement activity (appendix pp 2–4, 12–17). Small-scale community-based surveys were conducted in eight countries (ie, Australia, Bangladesh, Belgium, Brazil, Canada, Qatar, Singapore, and the UK; total N=161), interviews were done in Ghana (N=21), and a nationwide population-based panel survey was implemented in five countries (Kenya, Malaysia, South Africa, the UK, and the USA; total N=4995).

To explore what matters to people before pregnancy, participants in the small-scale community-based surveys and interviews were asked the hypothetical question "Imagine you were currently hoping to become pregnant (or for your partner to become pregnant); what would be the most important factors for you to have in place before pregnancy?". People were also asked to rank seven groups of predefined factors from most to least important to consider before pregnancy (ie, mental health, physical health, money, living conditions, relationships and family, work and

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Domain	Individual level	System level
Wider determinants of health	Ethnicity, First Nations or Indigenous status, education, health literacy in relation to pregnancy and parenting, employment, deprivation, and migrant health factors	Quality and accessibility of educational opportunities, assessment of working conditions conducive to health and wellbeing and security, availability of parental leave, availability of affordable housing, safe neighbourhoods, and access to private or public transport
Health care	Preconception assessment, routine physical and mental health checks, and routine dental appointments	Equitable access to services and programmes providing preconception care; access to health-care providers trained in preconception health; distance to nearest health-care facility; confidence in accessing relevant support services before, during, and after pregnancy; availability and enforcement of clinical preconception care guidelines; health insurance coverage
Emotional and social health and support	Domestic abuse, support network and social support (eg, from friends, family, and partner), family pressure (eg, having children or not, number of children, staying at home, or returning to work)	Availability and accessibility of support services for victims and survivors of domestic abuse
Reproductive health and family planning	Pregnancy intention, contraception, age, interpregnancy interval, obstetric history, and fertility issues	Effective contraceptive methods covered by insurance plans or available at no or reduced cost, access to family planning (ie, contraception) and sexual and reproductive health services, access to affordable assisted reproductive technology, safe and supportive abortion services and policies
Health behaviours and weight	Folic acid supplement use, other vitamin supplement use, vitamin deficiency, dietary intake, physical activity, sleep routine, smoking, second-hand smoke exposure, alcohol consumption, substance use, weight status (ie, BMI), and weight perception	Presence of policies mandating the fortification of staple foods (eg, flour or cereals) with folic acid, enforcement of age restrictions on alcohol and tobacco sales, food security and insecurity, and proximity, size, and quality of green spaces accessible to residents
Environmental exposures	Toxic or hazardous substances or exposures	Air pollution levels, safe drinking water, and sanitation and hygiene
Preventive health screening	Cervical screening	Availability and accessibility of effective screening methods
Immunisation and infections	Immunisations, vaccinations, infections (eg, malaria, hepatitis B, tuberculosis, and HIV), and sexually transmitted infections	Coverage rates for routine childhood vaccinations and seasonal influenza vaccines, provision of bednets in malaria-endemic areas
Mental health conditions	Mental health diagnosis, previous antenatal or postnatal mental illness, substance use treatment, and stress levels	Access to mental health facilities
Physical health conditions	Epilepsy, diabetes, polycystic ovary syndrome, endometriosis, thyroid disease, chronic hypertension, cardiovascular disease, previous thromboembolism, renal or kidney disease, asthma, rheumatological conditions, inflammatory bowel disease, sickle-cell disease or thalassaemia, phenylketonuria, developmental disability, cancer, female genital mutilation, pelvic floor dysfunction, and uterine anomalies	Access to annual, disease-specific reviews and check-ups
Medication	Use of prescription medication that is normally not recommended when planning pregnancy	Access to safe and relevant medication
Genetic risk	Personal or family history of a genetic disorder, previous pregnancy affected by an inherited genetic disorder, and consanguinity	Access to genetic screening

The individual-level and system-level indicators included are examples based on previous publications,^{16,19,20,28–36} community-based surveys conducted in eight countries (appendix pp 5–9), and input from coauthors. This list will be used as part of the international Core Indicators for Preconception Health and Equity Alliance's international Delphi study to reach consensus on a core set of preconception health surveillance indicators that can be used or adapted for use across low-income, middle-income, and high-income countries.

Table 1: Examples of indicators of pregnancy planning and preconception health at the individual and system level

education, and climate change), and to describe why some topics were more important than others. Questions and ranking measures were not validated or tested for reliability.

Most important factors to have in place before pregnancy

The most frequently mentioned topics were about health (including wellbeing and mental health), financial stability, and support (including partner and family relationships; figure 3; appendix pp 5–9). Consistent with the survey findings, interviews in Ghana identified socioeconomic and health-related factors as two sub-themes of the most important factors to consider before pregnancy (panel). When community-based survey participants were asked to rank seven preconception topics, mental health was generally rated as the most important factor, closely followed by physical health (appendix pp 10–11; table 2). Relationships and family,

money, and living conditions were generally higher than work and education, and climate change.

Findings from the panel survey differed slightly, but confirmed physical health, mental health, and relationships and family as factors that people prioritise when hoping to become pregnant (appendix p 11; table 3). Differences between women and men were small, with women tending to place slightly higher importance on relationships and family and mental health, whereas men more often emphasised the importance of money and work and education across most countries.

When asked how they decided which factors to prioritise, community members focused on: what would be most important for their child (or future child), what they were currently struggling with (eg, mental health or finding an affordable house), and their personal experience (eg, previous pregnancies, their own childhood, or the experiences of family and friends).

	Overall	Australia	Bangladesh	Belgium	Brazil	Canada	Qatar	Singapore	UK
Seven groups of factors	n=161	n=15	n=6	n=39	n=36	n=10	n=16	n=22	n=17
Mental health	917*	83	36*	222*	195*	56*	97*	122*	106*
Physical health	828	85*	33	189	171	45	89	122*	94
Relationships and family	705	76	25	143	169	43	67	107	75
Money	683	63	16	187	154	39	63	92	69
Living conditions	694	68	32	175	156	56	68	81	58
Work and education	455	38	15	112	116	19	42	71	42
Climate change	238	16	11	65	47	22	22	22	33

Values are borda counts (the frequency of item selected per choice multiplied by weighting, eg, 7 for choice 1, and 1 for choice 7), with higher counts indicating higher priority. *Values for the highest priorities.

Table 2: Relative importance of seven groups of factors to have in place before pregnancy based on small-scale, community-based surveys

	Overall	Kenya		Malaysia		South Africa		UK		USA	
		Women	Men	Women	Men	Women	Men	Women	Men	Women	Men
Seven groups of factors	n=4995	n=520	n=480	n=520	n=480	n=510	n=490	n=527	n=468	n=520	n=480
Mental health	24 460	2805	2429	2590	2290	2448	2232	2486	2085	2626	2309
Physical health	27 653*	3066*	2701*	2998*	2653*	2787*	2610*	2850	2544*	2807*	2595*
Relationships and family	25 026	2396	2369	2444	2457	2503	2523	2887*	2332	2568	2411
Money	21 259	2021	1906	2416	2042	2165	2093	2201	1818	2200	2154
Living conditions	18 874	2023	1827	1925	1527	1974	1902	2178	1440	2123	1676
Work and education	14 140	1319	1398	1372	1397	1641	1606	2113	860	1460	1353
Climate change	8588	930	810	815	1074	762	754	811	743	776	931

Values are borda counts (the frequency of item selected per choice multiplied by weighting, eg, 7 for choice 1, and 1 for choice 7), with higher counts indicating higher priority. *Values for the highest priorities.

Table 3: Relative importance of seven groups of factors to have in place before pregnancy based on nationwide, population-based panel surveys

Independent organisations, such as the Guttmacher Institute and the Kaiser Family Foundation, have implemented forms of surveillance that are largely driven by organisational or policy goals. For example, following the 2023 *Dobbs v Jackson Women's Health Organization* ruling, which imposed stricter restrictions on access to abortion services in the USA, the impact of these policies on women's health has been the subject of increased focus. Moreover, the rising maternal mortality rates and the substantial inequalities in maternal mortality have also attracted attention. In response, the Enhancing Reviews and Surveillance to Eliminate Maternal Mortality Initiative was launched by the CDC in 2019 to identify areas for improvement and standardise measurement of key indicators.⁷² Furthermore, the USA Maternal Vulnerability Index was launched in 2021 as a tool to identify where and why mothers and birthing people are vulnerable to poor health outcomes to develop more precise, localised solutions.^{35,73}

Preconception health surveillance in England

Health, social, and demographic data in England are routinely collated by multiple organisations, including the UK Government's Office for Health Improvement and Disparities (OHID), the Office for National

Statistics, and the National Health Service. These data are available through several platforms, most comprehensively via OHID's Fingertips platform, which compiles data from multiple sources into topic-specific Public Health Profiles. Currently, however, no dedicated profile consolidates indicators on pregnancy planning and preconception health. Some relevant indicators are reported across other profiles—such as the Child and Maternal Health and Sexual and Reproductive Health profiles—or have been included in the first national report on women's preconception health,^{13,21} with plans to integrate these into OHID's Fingertips platform to create a more comprehensive surveillance system for preconception health. Collaboration between OHID and the UK Preconception Partnership has been crucial in advancing preconception health surveillance in England. A review of national and international guidelines, position statements, and policy reports identified 65 indicators for which data are routinely collected across national health datasets, surveys, or cohort studies.¹⁹ This review informed the first collaborative national report on women's preconception health, focusing on 32 indicator measures derived from antenatal booking appointment data.^{13,21} These findings are already influencing national policy initiatives aimed at addressing disparities in

For more on the **Fingertips platform** see <https://fingertips.phe.org.uk/>

For more on the **UK Preconception Partnership** see <https://www.ukpreconceptionpartnership.co.uk/>

For more on the **USA Maternal Vulnerability Index** see <https://mvi.surgoghealth.com/>

health-care access and adverse maternal and neonatal outcomes.⁷⁴

Preconception health surveillance in the Netherlands

In 2018, the Dutch Ministry of Health, Welfare and Sport launched Solid Start, the nationwide action programme, in response to scientific evidence of lasting consequences of a poor start in life and large socioeconomic inequalities in perinatal health.^{75,76} These inequalities were first visualised in 2007 through heatmaps that revealed substantial variations in perinatal mortality rates across different neighbourhoods in the city of Rotterdam.⁷⁷ The Rotterdam Generation R study revealed that these variations can be explained largely by the accumulation of individual risk factors in these areas.⁷⁸ The heatmaps served to highlight the urgency of the issue, compelling local—and later, national—governments to take action.⁷⁹

The Solid Start programme is designed to improve the health and development of children, with a focus on children born to vulnerable parents.⁷⁵ Preconception health is one of the three key themes in the programme, with health and social care professionals (including midwives, psychologists, and social workers) encouraged to ask all women of reproductive age whether they plan to become pregnant in the next year.⁸⁰ If the answer is yes, women are directed to the validated interactive questionnaire ZwangerWijzer, which identifies risk factors and offers personalised advice on how to reduce or mediate these.^{81,82} Approximately 60 000 individuals complete the interactive tool each year, providing valuable data to guide individual preconception care consultations and support monitoring of preconception health at the population level.

Heatmaps continue to have an important role in improving the Solid Start programme,⁷⁹ such as through the Perinatal Atlas and the Vulnerability Atlas, the online interactive surveillance tools. More detailed data on preconception health will become available when pseudonymised ZwangerWijzer data are linked with multiple data sources through the Data-Infrastructure for Parents and Children.⁸³ This research, in collaboration with the National Institute for Public Health and the Environment, will contribute to the development of a set of indicators to monitor preconception health and care in the Netherlands.

Other initiatives

Drawing on these frameworks, other countries are beginning to develop core indicator sets, identify relevant data sources, and produce annual reports. The Preconception Health Network Australia (PCHN) facilitated the co-development of a core indicator set.³³ An audit of state-based and territory-based perinatal databases and the Maternity Information Matrix revealed that many suitable indicators are captured in routinely collected data.³⁴ Important data gaps and inconsistencies need to be addressed, but the PCHN aim to produce a

first national report and evidence-based framework for ongoing monitoring of women's preconception health in Australia.³⁴

In Brazil, nationwide household survey data from 2013 and 2019 have been used to describe the prevalence of preconception health indicators in women of reproductive age by sociodemographic characteristics.⁸⁴ During this time, access to health services (ie, medical and dental consultations, and cervical screenings) improved, but the prevalence of indicators such as hypertension, alcohol use, and obesity increased. Consistent inequalities were observed, particularly by ethnicity, education level, and region. Although there are no plans for routine surveillance of preconception health in Brazil, these data show that internationally comparable indicators and relevant data sources exist, raising the possibility of future integration into international surveillance systems.

In Qatar and several other Middle Eastern countries, such as Saudi Arabia and Bahrain, mandatory premarital screening is implemented due to high consanguinity rates.^{85,86} This screening facilitates routine preconception care and allows for the collection of core preconception health indicator data. Although the specific conditions screened vary by country, all couples preparing for marriage are tested for common genetic disorders, infectious diseases, and other health conditions (eg, diabetes). Expanding this testing programme to align with clinical preconception care guidelines could help establish a comprehensive preconception care service model, support the monitoring of preconception health trends in women and men, and shape national health policy.

Future directions for national and international surveillance

Although frameworks for national surveillance of preconception health are being developed and implemented, important gaps remain that should be addressed to enhance and expand national and international surveillance.

Integration with existing frameworks and resources

Public health surveillance platforms are used in many countries,^{67,72,73,87} and by international organisations such as WHO.^{10,88} These global infrastructures can be used to collate data relevant to preconception health, although large gaps in routine data collection systems should be addressed.^{1,2} Nationwide surveys conducted in many countries, such as the Demographic and Health Surveys, capture information on women's reproductive health. Future updates to these surveys, and other routine data collection systems, could provide opportunities to measure indicators of pregnancy planning and preconception health. These updates should also consider social and biological determinants of paternal preconception health, which are often overlooked.⁸⁹ In addition, although this paper focuses on core indicators, monitoring of core

For more on the **Questionnaire** see <https://www.zwangerwijzer.nl>

For more on the **Perinatal Atlas** see <https://www.waarstaatjegemeente.nl/>

For more on the **Vulnerability Atlas** see <https://kwetsbaarheid.kansenkaart.nl/>

outcomes (ie, parental, pregnancy, birth, and child health outcomes) is also needed to assess the effect of changes in preconception health at a population level.

To date, surveillance initiatives have primarily used data from pregnant and postpartum women. This approach excludes male reproductive partners, as well as individuals who wish to avoid pregnancy, and those who might wish to become parents but are unable to conceive or experience miscarriage or stillbirth. Datasets more broadly relevant to people of reproductive age might already be available in many countries and should be explored to avoid duplicating effort and resources. To adequately capture important preconception health indicators, multiple data sources that collectively track individuals across their reproductive years might need to be combined (figure 2).

Developing internationally agreed core indicators

The number of potentially relevant indicators described in table 1 is too large to be included in regional, national, and international measurement and surveillance systems. To facilitate the development of a prioritised set (or subsets) of internationally agreed core indicators, we established the International Core Indicators for Preconception Health and Equity (iCIPHE) Alliance.⁹⁰ The Alliance includes academic, clinical, policy, and public stakeholders from low-income, middle-income, and high-income countries.

During the first hybrid workshop in September, 2022, we identified the rationale for developing internationally agreed core indicators. The Alliance strongly supports the routine reporting of preconception health through national data sources, and the development of national infrastructures to inform the creation of national reports and surveillance systems. Furthermore, cross-country comparison will enable direct comparison of these core indicators and any changes over time, which will facilitate learning on successful intervention strategies across countries. The development of core indicators will also increase awareness about the key determinants of preconception health among the public, health-care professionals, and policy makers. Clinical practice can also be improved by encouraging the routine recording of preconception health indicators that are covered in clinical preconception care guidelines. Another aim of the iCIPHE Alliance is to promote equity by including core indicators related to structural and systematic barriers to preconception health. Collectively, these preconception health indicators can be used to hold governments, public health agencies, and health-care organisations accountable for developing and delivering interventions and services that improve preconception health and reduce inequalities.

The iCIPHE Alliance is conducting an international Delphi study to reach consensus on core preconception health surveillance indicators that can be used or adapted for use across low-income, middle-income, and high-income countries.⁹⁰

Working in partnership

Collaboration among stakeholders is needed to develop and share clear guidance on the measurement and monitoring of indicators at regional, national, and global levels.^{1,66,91} Key stakeholders include public health agencies, government bodies, policy makers and programme implementers, non-governmental organisations, health and social care providers, health insurers, and advocacy groups. Engaging these end-users and community organisations from the start is essential in ensuring that the indicators are fit for purpose and can be adopted in measurement and monitoring systems.

The involvement of key stakeholders in the production of high-quality data on preconception health indicators can also help drive positive political and programmatic change. However, careful selection of indicators and advocacy for policy change are essential to prevent unintended negative political, legal, and social impacts, and to support human rights principles (eg, autonomy and reproductive justice). Areas of concern that might require particular attention include potential entrenchment of stereotyped gender norms, such as implicitly placing sole responsibility for preconception health on women, and the risk that policies could penalise women's health-related behaviours by framing these as threats to future pregnancy outcomes.^{1,92}

The iCIPHE Alliance will continue to engage its global, multidisciplinary membership to work towards the shared goal of measuring and monitoring preconception health. We invite experts interested in preconception health surveillance to join the Alliance and participate in the Delphi study and subsequent dissemination and implementation activities. As the notion of monitoring preconception health gains traction, so does the need to research the indicators and concepts on which monitoring is based.

Cost and benefit considerations

Measuring surveillance indicators of preconception health involves specific costs and benefit considerations.^{93,94} On the cost side, resources are likely to be required for indicator development, data collection tools, training on consistent reporting, data management systems, continuous analysis and reporting, and regular reviews and updates of indicators. The benefits, although often challenging to quantify financially, might be substantial. They include gaining a comprehensive view of population preconception health, early detection of emerging trends, monitoring of health inequalities, and creation of an evidence base for programme development and policy evaluation. Also, active surveillance and response will bring about cost savings from cases prevented (eg, adverse birth outcomes). Standard measures, such as health-related quality of life and quality-adjusted life-years, can be used to compare disease burden.⁹⁵ Such surveillance data enables more informed decision making and targeted resource allocation in public health.

Although cost–benefit and return-on-investment (ROI) analyses are valuable for highlighting the economic value of preconception health interventions, such analyses are methodologically complex and rarely applied to surveillance systems at the outset.⁹⁶ Economic attributes aimed at assessing the value of surveillance systems (eg, their cost or effectiveness) have not often been considered for several reasons:⁹⁷ the absence of established data, the challenge of defining appropriate comparators, and the broad range of benefits (including the value of health information)—many of which are societal or long lasting, and difficult to quantify in monetary terms.

A business case might need to be developed when updating an existing or proposing a new surveillance system and should include an outline of the proposed system and its key benefits, background on the population's current preconception health status, and potential public health impacts. The proposal should also describe the system's features, how it will integrate with existing health information systems, and new or existing data to be collected and included. Risks and mitigation strategies, an implementation timeline, and an evaluation plan with key performance indicators, alongside a cost–benefit analysis providing an ROI metric, would further support the business case.

Although the business case should include an ROI metric for the surveillance system itself, comprehensive monitoring also enables more accurate ROI analyses of the preconception health interventions being evaluated, creating the dual benefit of cost-effective surveillance that facilitates cost-effective interventions.

Conclusion

As large-scale initiatives designed to improve preconception health are emerging in different populations and settings, systems that can monitor progress are essential for effective planning, advocacy, improving clinical practice, advancing equity, accountability, and use of resources. Discovering which aspects of preconception health matter most to people (eg, mental health) provides an important guide for developing future surveillance systems. Including system-level factors that hinder or support people to improve their health before and between pregnancies is equally important. In practical terms, this means combining relevant data collected routinely (for any purpose) from different sources into existing surveillance infrastructures. A few countries are on track to achieve this, and strong international collaboration is needed to achieve consensus on core indicators that can be compared across low-income, middle-income, and high-income countries.

Contributors

DS and JS conceptualised the idea and outline for the Review. DS, JH, and JS conceptualised data collection on community views, coordinated by DS. DS, EAN, WVN, GFA-J, HFAR, NSK, ALVB, DM, CWK, JKYC, ID, and SAN led data collection on community views in their respective countries. DS and JH conducted template analysis and DS analysed ranking data based on community survey data. JH and GB led the

writing of the section on individual-level indicators of pregnancy planning. SV led the writing of the section on system-level indicators and preconception health surveillance in the USA. ES led the writing of the section on preconception health surveillance in the Netherlands. GC led the writing of the section on cost and benefit considerations. DS created the figures with input from all authors. All authors provided input on the content before DS led the writing of the first full draft. All authors contributed to review and editing for important intellectual content.

Declaration of interests

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